



Higher-Order Derivative Sampling Associated with Fractional Fourier Transform

Rui-Meng Jing¹ · Qiang Feng^{1,2} · Bing-Zhao Li¹ 

Received: 27 July 2017 / Revised: 27 August 2018 / Accepted: 29 August 2018 /

Published online: 3 September 2018

© Springer Science+Business Media, LLC, part of Springer Nature 2018

Abstract

The uniform and recurrent nonuniform higher-order derivative sampling problems associated with the fractional Fourier transform are investigated in this paper. The reconstruction formulas of a bandlimited signal from the uniform and recurrent nonuniform derivative sampling points are obtained. It is shown that if a bandlimited function $f(t)$ has $n - 1$ order derivative in fractional Fourier transform domain, then $f(t)$ is determined by its uniform sampling points $f^{(l)}(knT)$ ($l = 0, 1, \dots, n - 1$) or recurrent nonuniform sampling points $f^{(l)}(n(t_p + kNT))$ ($l = 0, 1, \dots, n - 1$; $p = 1, 2, \dots, N$), the related sampling rate is also reduced by n times. The examples and simulations are also performed to verify the derived results.

Keywords Derivative sampling · Fractional Fourier transform · Uniform sampling · Recurrent nonuniform sampling

1 Introduction

The well-known Shannon sampling theorem [22,26] shows that when the sampling rate is larger than the Nyquist rate, the bandlimited signals can be reconstructed without

This work is supported by the National Natural Science Foundation of China (No. 61671063) and (No. 61861044).

✉ Bing-Zhao Li
li_bingzhao@bit.edu.cn

Rui-Meng Jing
1215021375@qq.com

Qiang Feng
fqtxingzhen@163.com

¹ School of Mathematics and Statistics, Beijing Institute of Technology Beijing, No. 5 Zhongguancun South Street, Haidian District, Beijing 100081, China

² School of Mathematics and Computer Science, Yanan University, Yanan 716000, Shanxi, China

distortion by their uniformly sampled signals. However, in most of the real applications, Shannon sampling conditions are not satisfied, and therefore the research of the sampling theory becomes one of the hottest research topics in modern signal processing and applied mathematical community. There have been many references about the generalizations of the classical Shannon sampling [3,5,8,10,11,15,17,21,33] with certain situations, among them, how to reduce the sampling rate of a signal by multi-channel sampling [25] or derivative sampling is one of the challenge problems in the sampling field.

The earliest works on sampling with derivative values in Fourier transform domain of a bandlimited function are proposed by Fogel [8], Jagerman and Fogel [11], and Linden and Ahramon [15]. In these works, Shannon sampling theorem is extended to uniform sampling of a bandlimited function $f(t)$ and their derivative $f^{(l)}(knT)$ ($l = 0, 1, \dots, n-1$) in Fourier transform domain [21]. Since then, many works are also focusing on the sampling theory associated with the nonuniform sampling because it occurs in many applications due to imperfect timebase or random events [3,5,10,17,33]. One of the most important types of nonuniform sampling is recurrent nonuniform sampling, the perfect reconstruction formula of recurrent nonuniform sampling is derived in [10,33]. It has been shown in [21] that a function $f(t)$ can be perfectly reconstructed by the recurrent nonuniform sampling points from its derivatives $f^{(l)}(n(t_p + kNT))$ ($l = 0, 1, \dots, n-1$; $p = 1, 2, \dots, N$).

All of the above-mentioned sampling theories are related to the Fourier transform because Fourier transform [3] has excellent effect in dealing with stationary signals whose statistical characteristics cannot change with time. However, we will not achieve the optimal results when dealing with the non-stationary signals if we still use the Fourier transform. Therefore, many other kinds of integral transforms, for example, the wavelet transform [4], the fractional Fourier transform (FrFT) [23,27], and linear canonical transform [2,9,18,32,35], are proposed to overcome these shortcomings. Among these novel transforms, the Fractional Fourier transform [23,27] is a new type of signal transformation and can deal with non-stationary signals, especially for chirp signals.

The classical Shannon uniform and nonuniform sampling theories have been well investigated and generalized to the FrFT in [12]. Along with the rich research results of sampling theories associated with the FrFT, the first-order uniform derivative sampling method is studied in [12,29] and the perfect reconstruction formula is obtained. Subsequently, the second-order uniform derivative sampling reconstruction formula is obtained in [13]. For the best of our knowledge, there has not been any paper published related to the much more higher-order uniform or nonuniform derivative sampling theorem in the FrFT domain. It is therefore interesting and worthwhile to carry out the research of higher-order derivative sampling associated with the FrFT.

In this paper, we investigate the uniform and recurrent nonuniform derivative sampling problems associated with FrFT. The perfect reconstruction formulas for a signal from the uniform or recurrent nonuniform sampling points of its higher derivatives are obtained, and the sampling rate can be reduced n times of the Nyquist sampling rate. A comparison and a quantitative analysis is provided to support our conclusions. The rest of the paper is organized as follows. Section 2 is preliminaries, the reconstruction formulas for uniform and recurrent nonuniform derivatives sampling are investigated

and obtained in Sect. 3. Section 4 provides the simulation results of the derived results. Subsequently, the error between reconstructed signal and original signal is analyzed by simulation results. Conclusions are given in Sect. 5.

2 Preliminaries

In order to represent our research results more clearly and easier to understand, some of the important results associated with FrFT and the derivative sampling in classical Fourier transform domain are reviewed in this section.

2.1 Fractional Fourier Transform (FrFT)

As a generalization of the classical Fourier transform, the α -th FrFT [16,20] of a signal $f(t)$, denoted as $F_\alpha(u)$, is defined as:

$$F_\alpha(u) = \mathcal{F}^\alpha[f(t)](u) = \int_{-\infty}^{+\infty} f(t) K_\alpha(u, t) dt \quad (1)$$

where the kernel function $K_\alpha(u, t)$ is given as follows:

$$K_\alpha(u, t) = \begin{cases} A_\alpha e^{j(t^2+u^2) \cot \alpha/2 - jtu \csc \alpha}, & \alpha \neq k\pi \\ \delta(t - u), & \alpha = 2k\pi \\ \delta(t + u), & \alpha = 2(k - 1)\pi \end{cases} \quad (2)$$

herein $A_\alpha = \sqrt{(1 - j \cot \alpha)/2\pi}$, $\mathcal{F}^\alpha$ is the FrFT operator with angle α . The FrFT domain at u axis is referred as the fractional domain.

The inverse FrFT is also a FrFT operation with parameter angle $-\alpha$, which can be expressed as follows:

$$f(t) = \mathcal{F}^{-\alpha}[F_\alpha(u)](t) = \int_{-\infty}^{+\infty} F_\alpha(u) K_\alpha^*(u, t) du \quad (3)$$

It is well known that the FrFT can be seen as the generalization of the classical Fourier transform, most of the well-known concepts and theorems associated with the Fourier transform are all investigated in the FrFT domain. For example, the sampling theories, the uncertainty principles [24,28,31] the Wigner–Ville distribution [1] and the bandlimited signal are discussed in details.

In classical Fourier transform domain, a function $f(t)$ is called a σ bandlimited if its energy is finite and its Fourier transform $G(\omega)$ vanishes outside the interval $(-\sigma, \sigma)$ in transform domain. That is to say, $F(\omega) = 0$ for $|\omega| \geq \sigma$,

$$\int_{-\infty}^{+\infty} |f(t)|^2 dt = (1/2\pi) \int_{-\sigma}^{+\sigma} |F(\omega)|^2 d\omega < \infty. \quad (4)$$

Similar with the definition of bandlimited signal in classical Fourier domain, the lowpass bandlimited definition of a signal related to the FrFT domain is proposed in [30], and the following relationship between the signal bandlimited in Fourier domain and FrFT domain has been derived. Suppose that a signal $f(t)$ is a Ω lowpass bandlimited signal in the FrFT domain, whose order is α . If we let

$$g(t) = \int_{-\Omega}^{\Omega} F^{\alpha}(u) e^{-ju^2 \cot \alpha/2 + jut \csc \alpha} du, \quad (5)$$

then, the signal $g(t)$ is a lowpass signal with the bandwidth $\frac{\Omega}{\csc \alpha}$ in the Fourier transform domain.

2.2 Generalized Sampling Model Associated with Fourier Transform

We review the generalized basic sampling model proposed in [4], given m linear systems with system functions $H_1(\omega), \dots, H_m(\omega)$; the sampling functions $x_i(t)$ of the original signal $g(t)$ can be given as follows:

$$x_i(t) = (1/2\pi) \int_{-\sigma}^{+\sigma} G(\omega) H_i(\omega) e^{j\omega t} d\omega \quad (6)$$

In this case, the sampling problem can be summarized as how to express $g(t)$ in terms of the sampling points $x_i(kmT)$ ($i = 1, 2, \dots, m$; $k = 0, \pm 1, \pm 2, \dots$) of these functions, where $T_0 = mT = 2\pi/c$, $c = 2\sigma/m$, and $x_i(kmT)$ is the sampling function values of $g(t)$. So sampling interval T_0 is m times of the original sampling interval T .

For this purpose, the following system is given [4]

$$H_1(\omega + rc)Y_1(\omega, t) + \dots + H_m(\omega + rc)Y_m(\omega, t) = e^{jrc t} \quad (7)$$

in which, $r = 0, 1, \dots, m-1$. This system defines m functions $Y_1(\omega, t), \dots, Y_m(\omega, t)$ of ω and t . Based on this system, the original bandlimited signal $g(t)$ can be perfectly reconstructed by the following formula [4]:

$$g(t) = \sum_{-\infty}^{+\infty} [x_1(kmT)y_1(t - kmT) + \dots + x_m(kmT)y_m(t - kmT)] \quad (8)$$

where

$$y_i(t) = (1/c) \int_{-\sigma}^{\sigma+rc} Y_i(u, t) e^{j\omega t} d\omega, \quad i = 1, 2, \dots, m. \quad (9)$$

2.3 Recurrent Nonuniform Sampling

In the recurrent nonuniform sampling model, each period consists of N nonuniform sampling points. Denoting the points in one period by t_p , $p = 1, 2, \dots, N$, the complete set of sampling points are $t_p + nNT$, $p = 1, 2, \dots, N$, $n \in (-\infty, +\infty)$. The model of this periodic nonuniform sampling distribution is shown in Fig. 1.

The sampling theories related to the recurrent nonuniform sampling in the classic Fourier domain are well investigated and discussed in [34]. Based on the research of the theories and applications of FrFT, the generalization of the classical recurrent nonuniform sampling theorem is derived in [12]. It is shown that if a continuous-time signal $f(t)$ which is a Ω band limited in the FrFT domain with order α , then it can be reconstructed from its nonuniform sampling points at a set of sampling times $t_n(t_n = t_p + nNT)$ [6,7,12,14,19]

$$f(t) = e^{-jt^2 \cot \alpha / 2} \sum_{n=-\infty}^{+\infty} \sum_{p=1}^N f(t_n) e^{\frac{j}{2}(t_n)^2 \cot \alpha} \psi_p(t)$$

$$\psi_p(t) = \frac{\prod_{q=1}^N \sin \frac{\Omega}{\sin \alpha N} (t - t_q)}{\prod_{\substack{q=1 \\ q \neq p}}^N \sin \frac{\Omega}{\sin \alpha N} (t_p - t_q)} \frac{(-1)^{nN}}{\frac{\Omega}{\sin \alpha N} (t - t_p - nN \frac{\pi \sin \alpha}{\Omega})} \quad (10)$$

3 Main Results

Based on the research results of FrFT and the derivative sampling, we investigate the higher-order derivatives sampling problems associated with the FrFT in this section. We firstly discuss the uniform derivative sampling and then investigate the recurrent nonuniform derivative sampling.

3.1 Uniform Derivative Sampling

Theorem 1 Suppose a signal $f(t)$ and its $n - 1$ order derivatives are continuous, and $f(t)$ is a bandlimited signal whose bandwidth is Ω in the FrFT domain of order α . Then the original signal $f(t)$ can be reconstructed from the sampling points of its derivatives by the following reconstruction formula. In this case, the sampling interval is n times of the original sampling interval T , that is to say, the sampling rate is reduced by n times.

$$f(t) = e^{-\frac{j}{2}t^2 \cot \alpha} \sum_{k=-\infty}^{+\infty} \sum_{l=0}^{n-1} \left(e^{\frac{j}{2}(knT)^2 \cot \alpha} f(knT) \right)^{(l)} s_l(t - nkT) \quad (11)$$

in which

$$s_l(t) = \sum_{r=l}^{n-1} a_{rl} \overline{s_r}(t), l = 0, 1, \dots, n-1, \quad (12)$$

$$\overline{s_r}(t) = \frac{1}{r!} t^r \operatorname{sinc}^n \left(\frac{t}{nT} \right), \quad (13)$$

and the coefficients a_{rl} are the solutions of

$$s_l^{(l')}(0) = \sum_{r=l}^{n-1} \overline{s_r}^{(l')}(0) a_{rl} = \delta_{ll'} \quad (14)$$

$$l' = l, \dots, n-1; l = 0, \dots, n-1.$$

Proof From the definition of the FrFT and the relationship between the bandlimited signal in Fourier and FrFT domain [9,12,32], we know that

$$f(t) = \int_{-\Omega}^{\Omega} F^{\alpha}(u) A_{\alpha} e^{-\frac{j}{2}(t^2+u^2) \cot \alpha + jut \csc \alpha} du = g(t) A_{\alpha} e^{-\frac{j}{2}t^2 \cot \alpha}$$

therefore

$$f(t) = \sqrt{\frac{1-j \cot \alpha}{2\pi}} e^{-\frac{j}{2}t^2 \cot \alpha} g(t), \quad (15)$$

and $g(t)$ is a bandlimited signal with a bandwidth of $\frac{\Omega}{\csc \alpha}$ in Fourier transform domain. Applying the classical derivative sampling theorem derived in [17] to signal $g(t)$, we obtain

$$g(t) = \sum_{k=-\infty}^{+\infty} \sum_{l=0}^{n-1} g^{(l)}(knT) s_l(t - nkT) \quad (16)$$

where

$$s_l(t) = \sum_{r=l}^{n-1} a_{rl} \overline{s_r}(t), l = 0, \dots, n-1$$

$$\overline{s_r}(t) = \frac{1}{r!} t^r \operatorname{sinc}^n \left(\frac{t}{nT} \right)$$

and the coefficients a_{rl} are the solutions of

$$s_l^{(l')}(0) = \sum_{r=l}^{n-1} (0)a_{rl} = \delta_{ll'}, l' = l, \dots, n-1; l = 0, \dots, n-1.$$

According to Eq. (15), we derive

$$g^{(l)}(knT) = \left(\sqrt{\frac{2\pi}{1-j\cot\alpha}} e^{\frac{j}{2}t^2\cot\alpha} f(t) \right)^{(l)} \Big|_{t=knT} \quad (17)$$

By substituting Eqs. (16) and (17) into Eq. (15), we can obtain the final conclusion. $\square$

Theorem 1 shows that a bandlimited signal can be perfectly reconstructed from its sampling points of derivatives, and this will be greatly helpful for simplification of derivative sampling in FrFT domain. Theorem 2 gives another representation of higher-order derivative sampling of a bandlimited signal in FrFT domain.

Theorem 2 Assuming that $f(t)$ is a fractional band limited signal satisfying the conditions of Theorem 1, then $f(t)$ can be expressed as the following formula,

$$f(t) = e^{-\frac{j}{2}t^2\cot\alpha} \sum_{k=-\infty}^{+\infty} \sum_{l=0}^{n-1} \left(e^{\frac{j}{2}(knT)^2\cot\alpha} f(knT) \right)^{(l)} y_{l+1}(t - knT) \quad (18)$$

where $y_{l+1}(t)$ meets the conditions of Eq. (9).

Proof From the definition of the FrFT and the relationship between the bandlimited signal in the Fourier and the FrFT domain, we know that

$$f(t) = \sqrt{\frac{1-j\cot\alpha}{2\pi}} e^{-\frac{j}{2}t^2\cot\alpha} g(t) \quad (19)$$

$$g(t) = \sum_{k=-\infty}^{+\infty} x_1(knT)y_1(t - knT) + \dots + x_n(xnT)y_m(t - knT) \quad (20)$$

where $g(t)$ is a bandlimited signal with a bandwidth of $\frac{\Omega}{\csc\alpha}$ in Fourier transform domain, $x_i(knT)$ are sampling functions of $g(t)$. Let $H_i(\omega) = (j\omega)^{i-1}$, then $x_i(t) = g^{(i-1)}(t)$. Let $i-1 = l$, $i = 1, \dots, m$ and $m = n$, therefore $l = 0, 1, \dots, n-1$, and

$$g(t) = \sum_{k=-\infty}^{+\infty} \sum_{l=0}^{n-1} g^{(l)}(knT)y_{l+1}(t - knT) \quad (21)$$

$$g^{(l)}(knT) = \left(\sqrt{\frac{2\pi}{1-j\cot\alpha}} e^{\frac{j}{2}t^2\cot\alpha} f(t) \right)^{(l)} \Big|_{t=knT} \quad (22)$$

By substituting Eqs. (21) and (22) into Eq. (19), we can draw the final conclusion. $\square$

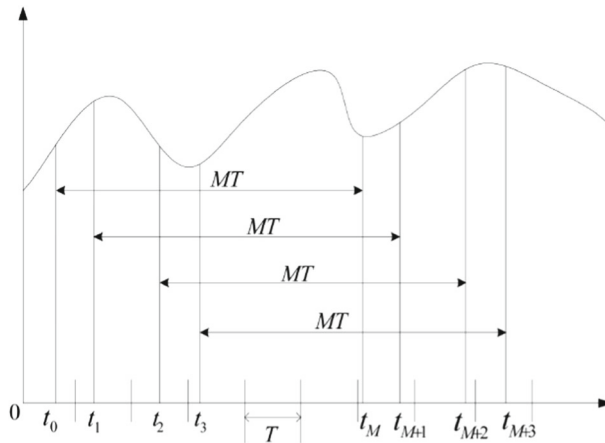


Fig. 1 Periodic nonuniform sampling model ($M = N$) [5]

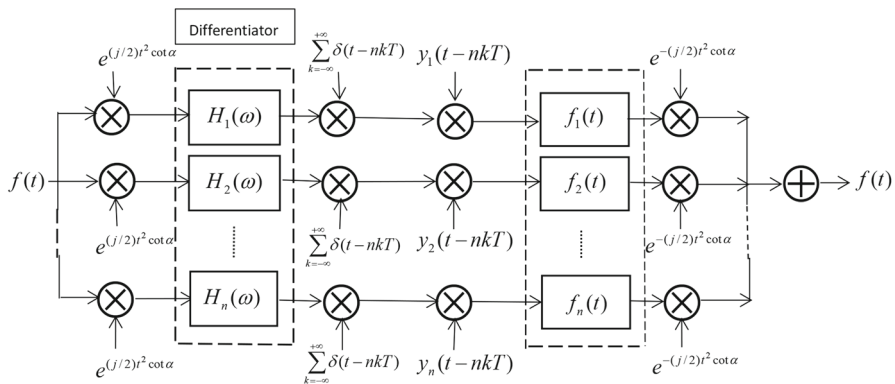


Fig. 2 Interpolation and reconstruction of a signal and its derivative

Theorem 2 gives another representation of the multi-channel derivative sampling associated with the FrFT, this process can be interpreted by using a block diagram, as shown in Fig. 2. From the point of view of computational complexity, it is easy to check that the complexity of the above formula and the lower-order derivative formula derived in [12] is $O(N^2)$. However, the sampling rate of the higher-order derivative sampling is lower than that of the lower derivative, and which can be used in broadband signal processing situations [9,32].

3.2 Recurrent Nonuniform Derivative Sampling

Theorem 3 Suppose $f(t)$ is sampled in accordance with the recurrent nonuniform model in Sect. 2.3, and $f(t)$ is a fractional bandlimited signal satisfying the conditions in Theorem 1, then $f(t)$ can be expressed as

$$f(t) = e^{-\frac{j}{2}t^2 \cot \alpha} \sum_{k=-\infty}^{+\infty} \sum_{p=1}^N \sum_{l=0}^{n-1} \left(e^{\frac{j}{2}(n(t_p+kNT))^2 \cot \alpha} f(n(t_p+kNT)) \right)^{(l)} \cdot \overline{s_{lp}}[t - n(t_p+kNT)]$$

in which,

$$\overline{s_{lp}}(t) = \sum_{r=l}^{n-1} a_{rp} \overline{\overline{s_{rp}}}(t), l = 0, \dots, n-1, p = 1, \dots, N \quad (23)$$

$$\overline{\overline{s_{rp}}}(t) = \frac{1}{r!} t^r \left[\frac{\prod_{q=1}^N \sin \pi((t+n(t_p-t_q))/nNT)}{\prod_{q \neq p}^N \sin \pi(n(t_p-t_q)/nNT)} \frac{1}{\pi(\frac{t}{nNT})} \right]^n \quad (24)$$

and the coefficients a_{rp} are solutions of

$$\overline{s_{lp}}^{(l')}(0) = \sum_{r=l}^{n-1} a_{rp} \overline{\overline{s_{rp}}}^{(l')}(0) = \delta_{ll'}, l' = l, \dots, n-1; p = 1, \dots, N. \quad (25)$$

Proof From the definition of the FrFT and the relationship between the bandlimited signal in the Fourier and the FrFT domain, we know that

$$f(t) = \sqrt{\frac{1-j \cot \alpha}{2\pi}} e^{-\frac{j}{2}t^2 \cot \alpha} g(t) \quad (26)$$

and $g(t)$ is a band limited signal with a bandwidth of $\frac{\Omega}{\csc \alpha}$ in the Fourier domain. Applying the classical derivative sampling theorem derived in [17] to signal $g(t)$, we obtain

$$g(t) = \sum_{k=-\infty}^{+\infty} \sum_{l=0}^{n-1} g^{(l)}(knT) \overline{s_{lp}}(t - n(t_p+kNT)) \quad (27)$$

where

$$\overline{s_{lp}}(t) = \sum_{r=l}^{n-1} a_{rp} \overline{\overline{s_{rp}}}(t), l = 0, \dots, n-1, p = 1, \dots, N$$

$$\overline{\overline{s_{rp}}}(t) = \frac{1}{r!} t^r \left[\frac{\prod_{q=1}^N \sin \pi((t+n(t_p-t_q))/nNT)}{\prod_{q \neq p}^N \sin \pi(n(t_p-t_q)/nNT)} \frac{1}{\pi(\frac{t}{nNT})} \right]^n$$

and the coefficients a_{rl} are the solutions of

$$\overline{s}_{lp}^{(l')}(0) = \sum_{r=l}^{n-1} a_{rp} \overline{s}_{rp}^{(l')}(0), l' = l, \dots, n-1; p = 1, \dots, N.$$

According to Eq. (15), we derive

$$g^{(l)}(knT) = \left(\sqrt{\frac{2\pi}{1-j\cot\alpha}} e^{\frac{j}{2}t^2\cot\alpha} f(t) \right)^{(l)} \Big|_{t=knT} \quad (28)$$

By substituting Eqs. (27) and (28) into Eq. (26), we can draw the final conclusion. $\square$

The computational complexity of the above formula and the lower-order derivative formula in [30] is $O(M^2)$. However, using higher-order derivative periodic nonuniform sampling, we can collect and reconstruct the original information with the sampling rate which is much lower than the Nyquist sampling theorem. Therefore, in the case of the same computational complexity, the sampling rate of the higher-order derivative sampling is lower than that of the lower derivative, and the results derived in this paper can be used in broadband spectrum sensing.

4 The Simulations and Examples

In this section, some simulations and examples are discussed in detail to verify the correctness of the derived results. In this section, we suppose that

$$f(t) = \mathcal{F}^{-\alpha}[F_{\alpha}(u)](t) = \int_{-\infty}^{+\infty} F_{\alpha}(u) K_{\alpha}^*(u, t) du,$$

and let $F_{\alpha}(u) = 1$. So we obtain

$$\begin{aligned} f(t) &= \int_{-\infty}^{+\infty} K_{\alpha}^*(u, t) du \\ &= \int_{-\infty}^{+\infty} \sqrt{\frac{1+j\cot\alpha}{2\pi}} e^{-\frac{j}{2}(t^2+u^2)+jtu\csc\alpha} du \end{aligned} \quad (29)$$

Figure 3a shows a rectangular signal whose bandwidth is $\Omega = 4$ in the FrFT domain; Fig. 3b shows the absolute value of this rectangular signal in time domain, denoted as $|f(t)|$.

4.1 Uniform Derivative Sampling

In this case, we let $n = 4$ and figure out the reconstruction formula according to Theorem 1 of this paper. Based on Eqs. (13) and (14), we have worked out for the values of a_{rl} , $l = 0, 1, 2, 3$; $r = l, \dots, 3$ as following.

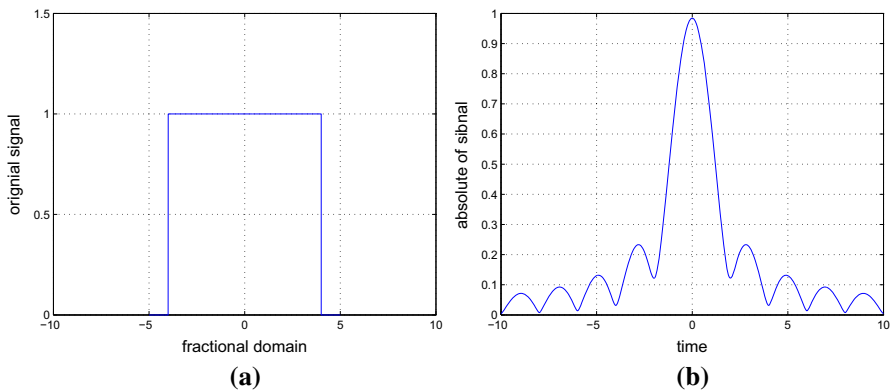


Fig. 3 Simulation results: **a** a bandlimited signal (rectangular signal) whose bandwidth is $\Omega = 4$ in the FrFT domain; **b** the absolute value of the rectangular signal in time domain

$$a_{00} = 1; a_{10} = 0; a_{20} = \frac{4\pi^2}{3}; a_{30} = 0;$$

$$a_{11} = 1; a_{21} = 0; a_{31} = 4\pi^2; a_{22} = 1; a_{32} = 0; a_{33} = 1.$$

By substituting these values of a_{rl} to Eq. (12), we can obtain the expression of $s_l(t)$, $l = 0, 1, 2, 3$.

$$s_0(t) = \left(1 + \frac{2}{3}\pi^2 t^2\right) \text{sinc}^4\left(\frac{t}{4T}\right) \quad (30)$$

$$s_1(t) = \left(1 + \frac{2}{3}\pi^2 t^2\right) t \text{sinc}^4\left(\frac{t}{4T}\right) \quad (31)$$

$$s_2(t) = \frac{1}{2} t^2 \text{sinc}^4\left(\frac{t}{4T}\right) \quad (32)$$

$$s_3(t) = \frac{1}{6} t^3 \text{sinc}^4\left(\frac{t}{4T}\right) \quad (33)$$

And according to Eq. (17), we obtain $g^{(l)}(t)$, $l = 0, 1, 2, 3$.

$$g(t) = \sqrt{\frac{2\pi}{1 - j \cot \alpha}} e^{\frac{j}{2} t^2 \cot \alpha} f(t) \quad (34)$$

$$g'(t) = \sqrt{\frac{2\pi}{1 - j \cot \alpha}} e^{\frac{j}{2} t^2 \cot \alpha} (jt \cot \alpha f(t) + f'(t)) \quad (35)$$

$$g''(t) = \sqrt{\frac{2\pi}{1 - j \cot \alpha}} e^{\frac{j}{2} t^2 \cot \alpha} \left[((jt \cot \alpha)^2 + j \cot \alpha) f(t) + 2jt \cot \alpha f'(t) + f''(t) \right] \quad (36)$$

$$g'''(t) = \sqrt{\frac{2\pi}{1-j\cot\alpha}} e^{\frac{j}{2}t^2\cot\alpha} [(jt\cot\alpha)^3 + 3(j\cot\alpha)^2t)f(t) + (3(jt\cot\alpha)^2 + 3j\cot\alpha)f'(t) + 3jt\cot\alpha f''(t) + f'''(t)] \quad (37)$$

Based on the result of Theorem 1 and the above equations, we can obtain the reconstruction formula as following.

$$f(t) = e^{-\frac{j}{2}t^2\cot\alpha} \sum_{k=-\infty}^{+\infty} e^{\frac{j}{2}(4kT)^2\cot\alpha} \left[(1 + j4kT\cot\alpha(t - 4kT) + \left(\frac{2}{3}\pi^2 + \frac{1}{2}(j4kT\cot\alpha)^2 + \frac{1}{2}j\cot\alpha \cdot (t - 4kT)^2 + \left(\frac{2}{3}\pi^2 j4kT\cot\alpha + \frac{1}{6}(j4kT\cot\alpha)^3 + \frac{1}{2}(j\cot\alpha)^2 4kT\right)(t - 4kT)^3 \cdot f(4kT) + ((t - 4kT) + j4kT\cot\alpha(t - 4kT)^2 + \left(\frac{2}{3}\pi^2 + \frac{1}{2}j\cot\alpha + \frac{1}{2}(j4kT\cot\alpha)^2)(t - 4kT)^3\right) \cdot f'(4kT) + \left(\frac{1}{2}(t - 4kT)^2 + \frac{1}{2}(j4kT\cot\alpha)(t - 4kT)^3\right) \cdot f''(4kT) + \frac{1}{6}(t - 4kT)^3 \cdot f'''(t - 4kT)) \right] \cdot \text{sinc}^4\left(\frac{t}{4T} - k\right). \quad (38)$$

In simulation, we assume that $f(t)$ is a FrFT domain bandlimited signal $\alpha = \arcsin(2/\pi)$. So the original sampling interval in time domain is $T = \pi \sin \alpha / \Omega = 0.5$, and sampling interval of derivative is $nT = 2$. That is to say, sampling rate of derivative is $\frac{1}{4}$ times the original sampling rate. By substituting $T = 0.5$, $\alpha = \arcsin(2/\pi)$ and Eq. (29) into Eq. (38), we can get the reconstruction formula. Figure 4a, b shows the real-part uniform sampling and imaginary part uniform sampling of $f(t)$, respectively.

Figure 4c, d shows the contrast of real part and imaginary part between reconstruction signal based on the FrFT and original signal $f(t)$, respectively. Because there is no infinite sampling point to cause errors, we use multiple sampling points as much as possible to reduce the error. Just as shown in Fig. 4c, d, the waveform of reconstructed signal is very similar to the original signal.

4.1.1 The Comparison Between FrFT and FT

From the results of [21], if we choose the higher-order derivative associated with Fourier transform for $f(t)$, the following expression can be obtained.

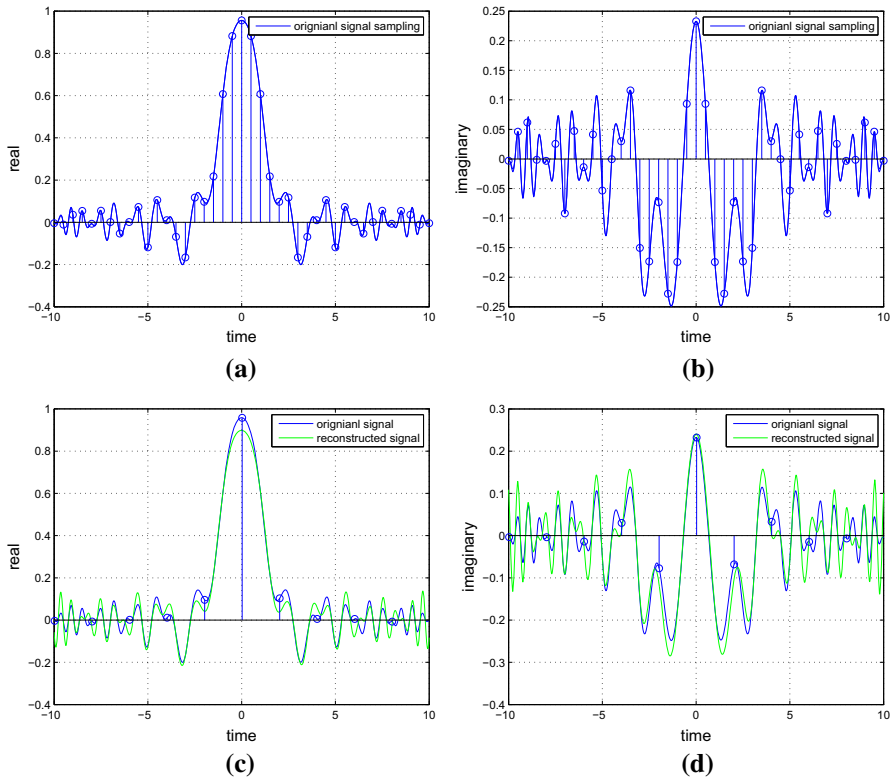


Fig. 4 Simulation results: uniform sampling **a** the real-part sampling of the signal in Fig. 3b; **b** the imaginary part sampling of the signal in Fig. 3b; **c** the contrast of real part between reconstruction signal and original signal; **d** the contrast of imaginary part between reconstruction signal and original signal

$$f(t) = \sum_{k=-\infty}^{+\infty} \left[\left(1 + \frac{2}{3} \pi^2 (t - 4kT)^2 \right) f(4kT) + \left(t - 4kT + \frac{2}{3} \pi^2 (t - 4kT)^3 \right) f'(4kT) \right. \\ \left. + \frac{1}{2} (t - 4kT)^2 f''(4kT) + \frac{1}{6} (t - 4kT)^3 f'''(4kT) \right] \text{sinc}^4 \left(\frac{t}{4T} - k \right) \quad (39)$$

Figure 5a, b plots the real part and imaginary part of reconstructed signal and original signal $f(t)$ based on FT, respectively. We can see that the reconstruction effect of Eq. (39) is far from Eq. (38). Through the simulation experiment, the mean square error of 2001 points in the real part of Fig. 4 is 3.0813, and the mean square error of 2001 points in the imaginary part of Fig. 4 is 2.6421. However, the mean square error of 2001 points in the real part of Fig. 5 is 70.8313, and the mean square error of 2001 points in the imaginary part of Fig. 5 is 16.5308. Figure 6a, b plots the mean square error (MSE) comparison between FrFT and FT. It is easy to see that higher-order derivative sampling associated with the FrFT is better than the FT in chirp signal sampling.

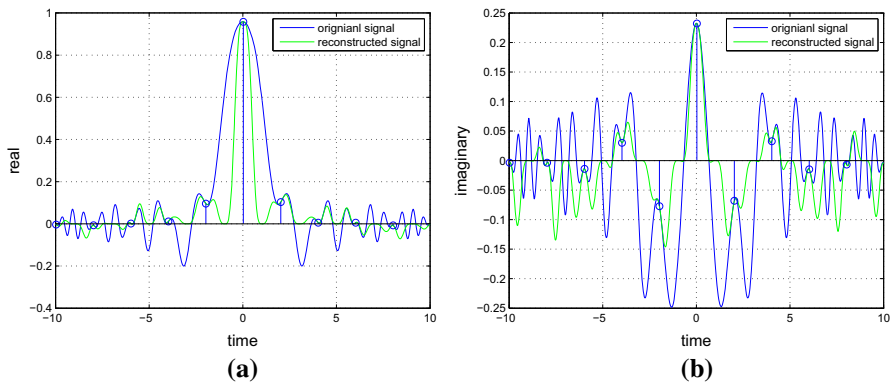


Fig. 5 Simulation results: uniform sampling **a** the contrast of real part between reconstruction signal based on Fourier transform and original signal; **b** the contrast of imaginary part between reconstruction signal based on Fourier transform and original signal

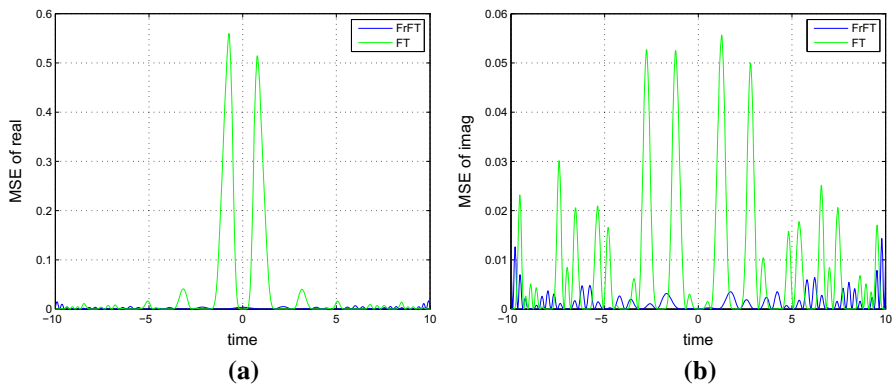


Fig. 6 Simulation results: uniform sampling **a** MSE of real part based on the fractional-order Fourier transform and Fourier transform; **b** MSE of image part based on the fractional-order Fourier transform and Fourier transform

4.1.2 The Comparison Between the Lower and Higher Order

From the results of [12], if we choose the lower-order derivative associated with FrFT for $f(t)$ and $n = 2$, the following expression can be obtained.

$$f(t) = e^{-\frac{j}{2}t^2 \cot \alpha} \sum_{k=-\infty}^{+\infty} e^{\frac{j}{2}(2kT)^2 \cot \alpha} [(1 + j2kT \cot \alpha(t - 2kT)) \cdot f(2kT) + (t - 2kT) \cdot f'(2kT)] \cdot \text{sinc}^2\left(\frac{t}{2T} - k\right). \quad (40)$$

Figure 7a, b plots the comparison between lower-order and higher-order derivative sampling of the reconstructed real part and imaginary part, respectively. We can see that the reconstruction effect of Eq. (40) is approximate to Eq. (38). Through the simulation

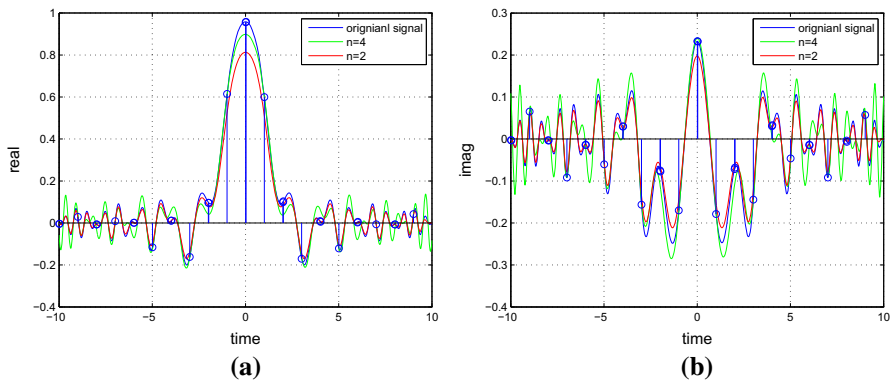


Fig. 7 Simulation results: uniform sampling **a** the contrast of real part; **b** the contrast of imaginary part

Table 1 Uniform derivative sampling (2001 sampling points)

Transformation type	MSE of real part	MSE of image part
Higher-order FT ($n = 4$)	70.8313	16.5308
Lower-order FrFT ($n = 2$)	3.9264	0.5505
Higher-order FrFT ($n = 4$)	3.0813	2.6421

experiment, we calculate that the mean square error in the real part of higher derivative sampling $n = 4$ is 3.0813, and the mean square error in the imaginary part of higher derivative sampling $n = 4$ is 2.6421. Moreover, the mean square error in the real part of low derivative sampling $n = 2$ is 3.9264, and the mean square error in the imaginary part of low derivative sampling $n = 2$ is 0.5505. The MSE comparison of uniform sampling is listed in following Table 1. In Table 1, the real-part reconstruction error of the higher-order derivative is smaller than that of the lower order, but the imaginary part is larger than the lower order. Although in terms of reconstruction error, higher derivatives have no obvious advantages over lower orders, they have a good advantage in terms of sampling rate. This method can be applied effectively when the sampling rate is required to be very low. And the error of reconstruction formula is what we need to study further.

A new signal shown in Fig. 8a, b represents the original signal in the FRFT domain and time domain. Figure 9a, b plots the comparison between lower-order and higher-order derivative sampling of the reconstructed real part and imaginary part, respectively. And the mean square error is shown in Table 2. The simulation results of the signal in Fig. 8 show that the error comparison is consistent with the results obtained in Table 1.

In addition, the computational complexity of the higher-order derivative sampling and the lower-order derivative sampling, both of which are $O((N = 2001)^2)$. According to the above analysis, the higher-order derivative sampling has lower sampling rate than the lower-order sampling, the former has more advantages in actual sampling.

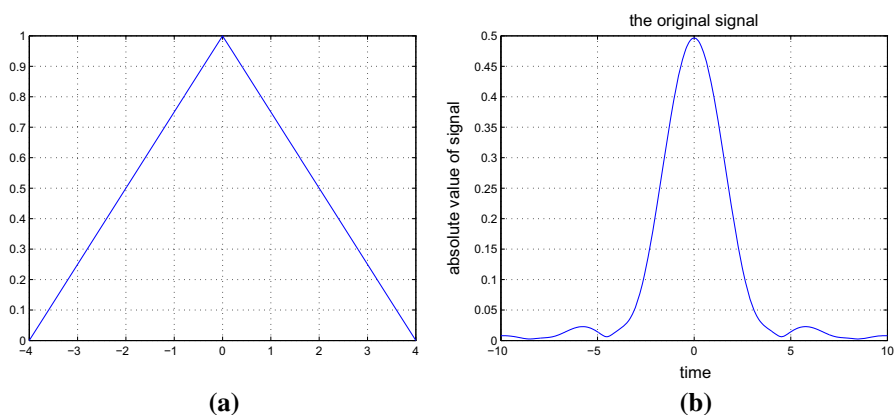


Fig. 8 Simulation results: uniform sampling **a** the original signal in the FrFT domain; **b** the absolute value of original signal in the time domain

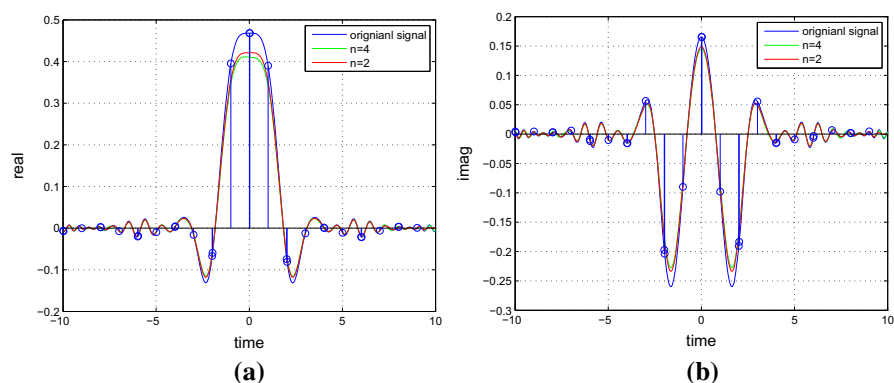


Fig. 9 Simulation results: uniform sampling **a** the contrast of real part; **b** the contrast of imaginary part

Table 2 Uniform derivative sampling (2001 sampling points)

Transformation type	MSE of real part	MSE of image part
Lower-order FrFT ($n = 2$)	3.8127	1.2101
Higher-order FrFT ($n = 4$)	3.5344	3.1372

4.2 Recurrent Nonuniform Derivative Sampling

As for nonuniform derivative sampling situations, if we let $n = 2$, $N = 3$, $\Omega = 2$, then we can figure out the reconstruction formula according to Theorem 3 derived in this paper.

Firstly, based on Eqs. (24) and (25), we work out the values of a_{rp} , $r = l, \dots, 1$; $l = 0, 1$; $p = 1, 2, 3$.

$$a_{0p} = 1; a_{1p} = 1; p = 1, 2, 3.$$

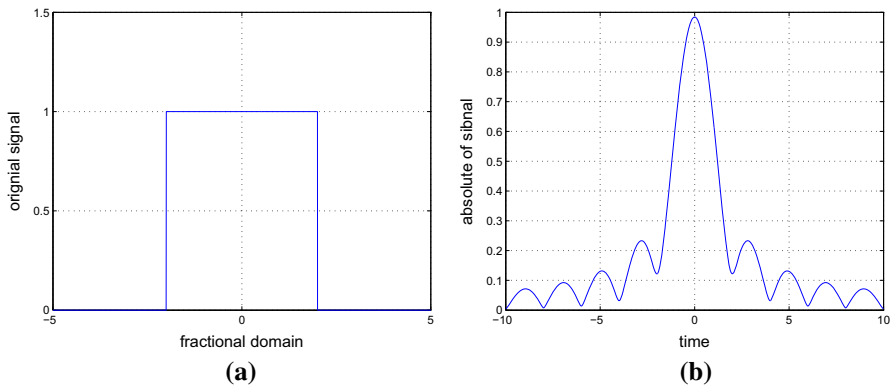


Fig. 10 Simulation results: **a** a band limited signal (rectangular signal) whose bandwidth is $\Omega = 2$ in FrFT domain; **b** the absolute value of the rectangular signal in time domain

Secondly, by bringing values of a_{rp} to Eq. (23), we can obtain the expression of $\overline{s_{lp}(t)}$, $l = 0, 1$; $p = 1, 2, 3$.

$$\overline{s_{0p}(t)} = (1 + t) \left[\frac{\prod_{q=1, q \neq p}^3 \sin \pi((t + 2(t_p - t_q))/6T)}{\prod_{q=1, q \neq p}^3 \sin \pi(2(t_p - t_q)/6T)} \frac{1}{\pi \frac{t}{6T}} \right]^2 \quad (41)$$

$$\overline{s_{1p}(t)} = t \left[\frac{\prod_{q=1, q \neq p}^3 \sin \pi((t + 2(t_p - t_q))/6T)}{\prod_{q=1, q \neq p}^3 \sin \pi(2(t_p - t_q)/6T)} \frac{1}{\pi \frac{t}{6T}} \right]^2 \quad (42)$$

Figure 10a shows a rectangular signal whose bandwidth is $\Omega = 2$ in the FrFT domain; Fig. 10b shows the absolute value of this rectangular signal in time domain, which is expressed as $|f(t)|$.

Thirdly, according to Eq. (17), we obtain $g^{(l)}(t)$, $l = 0, 1$.

$$g(t) = \sqrt{\frac{2\pi}{1 - j \cot \alpha}} e^{\frac{j}{2} t^2 \cot \alpha} f(t) \quad (43)$$

$$g'(t) = \sqrt{\frac{2\pi}{1 - j \cot \alpha}} e^{\frac{j}{2} t^2 \cot \alpha} (jt \cot \alpha f(t) + f'(t)) \quad (44)$$

Finally, based on the result of Theorem 3, we can obtain the desired reconstruction formula as following.

$$f(t) = e^{-\frac{j}{2} t^2 \cot \alpha} \sum_{k=-\infty}^{+\infty} \sum_{p=1}^{p=3} e^{\frac{j}{2} (2(t_p + 3kT))^2 \cot \alpha} f(2(t_p + 3kT))$$

$$(1 + (t - 2(t_p + 3kT))) \left[\frac{\prod_{q=1}^3 \sin \pi((t + 2(t_p - t_q))/6T)}{\prod_{q=1, q \neq p}^3 \sin \pi(2(t_p - t_q)/6T)} \frac{1}{\pi \frac{t}{6T}} \right]^2$$

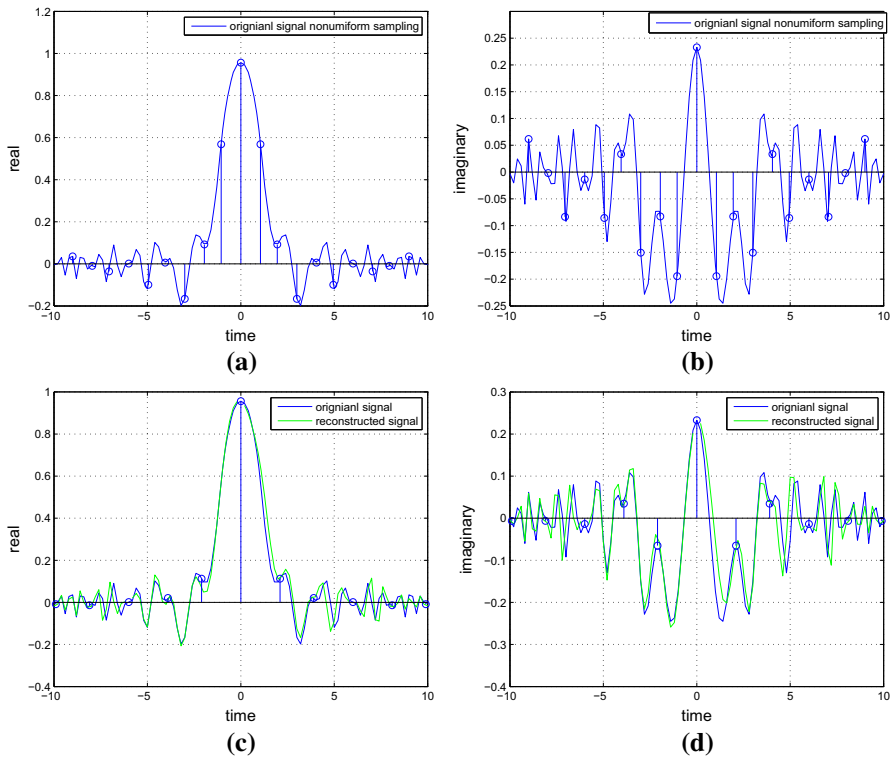


Fig. 11 Simulation results: recurrent nonuniform sampling **a** the real-part sampling of the signal in Fig. 8b; **b** the imaginary part sampling of the signal in Fig. 8b; **c** the contrast of real part between reconstruction signal and original signal; **d** the contrast of imaginary part between reconstruction signal and original signal

$$\begin{aligned}
 & + e^{\frac{j}{2}(2(t_p+3kT))^2 \cot \alpha} (2j(t_p+3kT) \cot \alpha f(2(t_p+3kT))) \\
 & + f'(2(t_p+3kT))(t-2(t_p+3kT)) \\
 & \cdot \left[\frac{\prod_{q=1}^3 \sin \pi((t+2(t_p-t_q))/6T)}{\prod_{q \neq p}^3 \sin \pi(2(t_p-t_q)/6T)} \frac{1}{\pi \frac{t}{6T}} \right]^2
 \end{aligned} \quad (45)$$

We assume that $f(t)$ is a FrFT domain bandlimited signal of $\Omega = 2$ and $\alpha = \arcsin(2/\pi)$, the reconstruction formula of uniform sampling when $n = 2$ and $N = 3$ can be derived. So the original sampling points in time domain are $t_p + kNT = t_p + 3 \cdot 1 \cdot k$; $p = 1, 2, 3$ and sampling points of derivative are $2(t_p + kNT) = 2(t_p + 3 \cdot 1 \cdot k)$; $p = 1, 2, 3$, that is to say, sampling rate of derivative is $\frac{1}{2}$ times the original sampling rate. And we assume that $t_1 = 0$, $t_2 = 1.05$, $t_3 = 1.95$. By substituting those values and Eqs. (29) into (44), the reconstruction formula can be derived.

Figure 11a, b plots the real part and imaginary part of recurrent nonuniform sampling of $f(t)$, respectively. Figure 11c, d plots the contrast of real part and imaginary

part between reconstruction signal and original signal $f(t)$, respectively. Because there is no infinite sampling point to cause errors, we use multiple sampling points as much as possible to reduce the error. As can be seen from Fig. 11c, d, the waveform of reconstructed signal is very similar to the original signal.

4.2.1 The Comparison Between FrFT and FT

If we choose the higher-order derivative associated Fourier transform for $f(t)$, the following expression can be obtained [21].

$$\begin{aligned}
 f(t) = & \sum_{k=-\infty}^{+\infty} \sum_{p=1}^{p=3} f(2(t_p + 3kT))(1 + (t - 2(t_p + 3kT))) \\
 & \cdot \left[\frac{\prod_{q=1}^3 \sin \pi((t + 2(t_p - t_q))/6T)}{\prod_{q=1, q \neq p}^3 \sin \pi(2(t_p - t_q)/6T)} \frac{1}{\pi \frac{t}{6T}} \right]^2 \\
 & + f'(2(t_p + 3kT))(t - 2(t_p + 3kT)) \\
 & \cdot \left[\frac{\prod_{q=1}^3 \sin \pi((t + 2(t_p - t_q))/6T)}{\prod_{q=1, q \neq p}^3 \sin \pi(2(t_p - t_q)/6T)} \frac{1}{\pi \frac{t}{6T}} \right]^2
 \end{aligned} \quad (46)$$

Figure 12a, b plots the contrast of real part and imaginary part between reconstruction signal and original signal $f(t)$ based on FT, respectively. We can see that the reconstruction effect of Eq. (46) is far from Eq. (45) from Figs. 11 and 12. Through the simulation experiment, the mean square error of 201 points in the real part of Fig. 11 is 0.2770, and the mean square error of 201 points in the imaginary part of Fig. 11 is 0.2527. However, the mean square error of 201 points in the real part of Fig. 12 is 1.1481, and the mean square error of 201 points in the imaginary part of Fig. 12 is 1.0661.

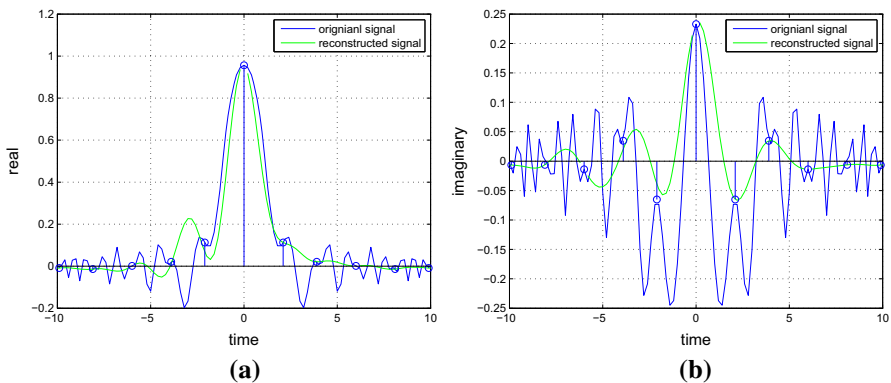


Fig. 12 Simulation results: periodic nonuniform sampling **a** the contrast of real part between reconstruction signal based on Fourier transform and original signal; **b** the contrast of imaginary part between reconstruction signal based on Fourier transform and original signal

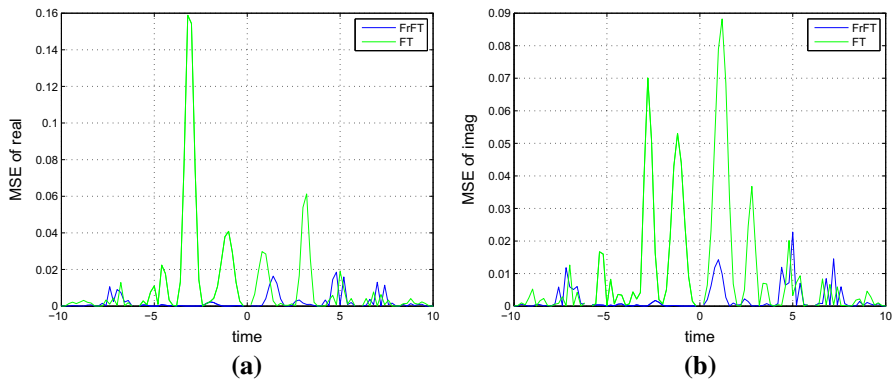


Fig. 13 Simulation results: periodic nonuniform sampling **a** MSE of real part based on the fractional-order Fourier transform and Fourier transform; **b** MSE of image part based on the fractional-order Fourier transform and Fourier transform

In order to see the differences more clearly, Fig. 13a, b plots the MSE of each point based on the FrFT and FT. So through this quantitative comparison, we can obtain that higher-order derivative sampling has the same performance as the traditional sampling rate, and it has better results in dealing with chirp class signals.

4.2.2 The Comparison Between Higher- and Lower-Order Derivative Sampling

From the results of [12], if we choose the lower-order derivative associated with FrFT for $f(t)$ and $n = 1$, the following expression can be obtained.

$$f(t) = e^{-\frac{j}{2}t^2 \cot \alpha} \sum_{k=-\infty}^{+\infty} \sum_{p=1}^{p=3} e^{\frac{j}{2}(t_p+3kT)^2 \cot \alpha} f(t_p + 3kT) \left[\frac{\prod_{q=1}^3 \sin \pi((t + t_p - t_q)/3T)}{\prod_{\substack{q=1 \\ q \neq p}}^3 \sin \pi(t_p - t_q/3T)} \frac{1}{\pi \frac{t}{3T}} \right] \quad (47)$$

Figure 14a, b plots the real part and imaginary part between lower-order and higher-order derivative sampling based on FrFT, respectively. We can see that the reconstruction effect of Eq. (47) is approximate to Eq. (45). Through the simulation experiment, the mean square error in the real part of higher derivative sampling $n = 2$ is 0.2270, and the mean square error in the imaginary part of higher derivative sampling $n = 2$ is 0.2527. Moreover, the mean square error in the real part of low derivative sampling $n = 1$ is 0.2620, and the mean square error in the imaginary part of low derivative sampling $n = 1$ is 0.2575. The MSE comparison of periodic nonuniform sampling is listed in Table 3.

In addition, the computational complexity of the higher-order derivative sampling and the lower-order derivative sampling is $O((N = 201)^2)$. According to the above analysis, it can be concluded that the higher-order derivative sampling has lower sampling rate than the lower-order sampling, the former has more advantages in actual

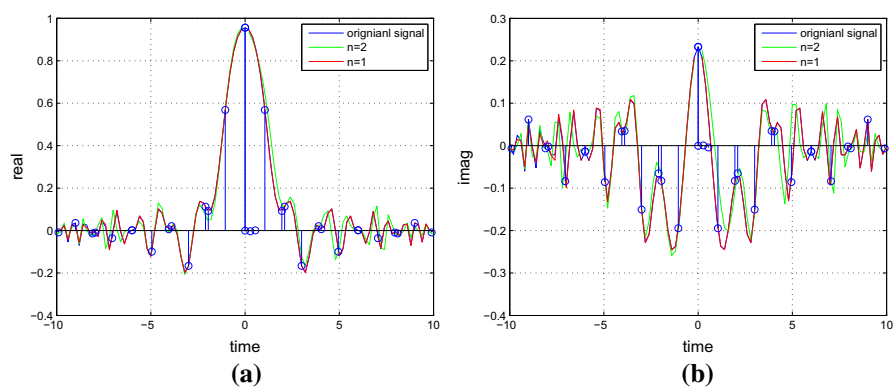


Fig. 14 Simulation results: uniform sampling **a** the contrast of real part; **b** the contrast of imaginary part

Table 3 Recurrent nonuniform derivative sampling (201 sampling points)

Transformation type	MSE of real part	MSE of image part
Higher-order FT ($n = 2$)	1.1481	1.0061
Lower-order FrFT ($n = 1$)	0.2620	0.2575
Higher-order FrFT ($n = 2$)	0.2270	0.2527

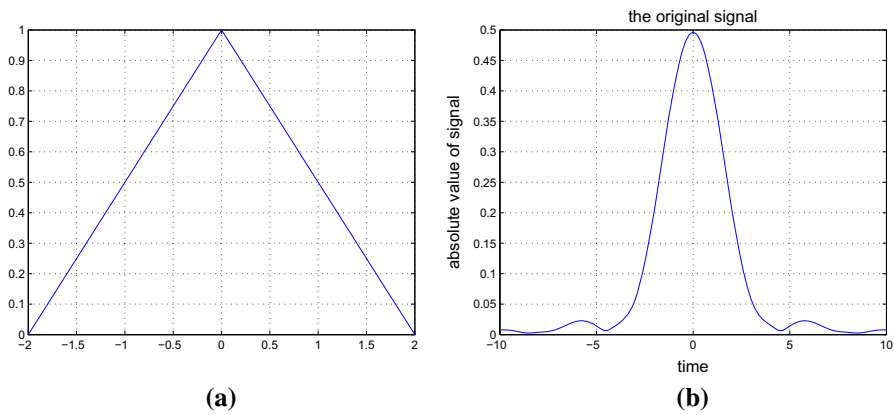


Fig. 15 Simulation results: nonuniform sampling **a** the original signal in the FrFT domain; **b** the absolute value of original signal in the time domain

sampling. In Table 3, we find that the real part and imaginary part reconstruction error of the higher-order derivative is smaller than that of the lower-order derivative, and next we will carry out the research of reconstruction error formula and analysis, in order to make to better use of our results.

A new signal shown in Fig. 15a, b represents the original signal in the FrFT domain and time domain. Figure 16a, b plots the comparison between lower-order and higher-order derivative sampling of the reconstructed real part and imaginary

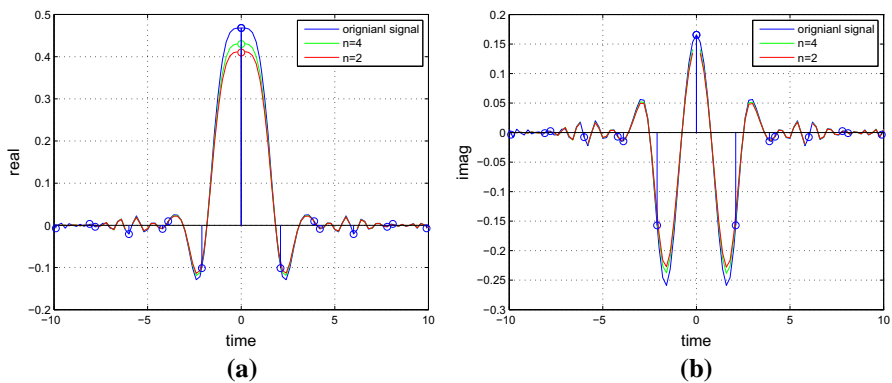


Fig. 16 Simulation results: nonuniform sampling **a** the contrast of real part; **b** the contrast of imaginary part

Table 4 Recurrent nonuniform derivative sampling (201 sampling points)

Transformation type	MSE of real part	MSE of image part
Lower-order FrFT ($n = 1$)	1.6111	0.4702
Higher-order FrFT ($n = 2$)	1.5606	0.9901

part, respectively. And the mean square error is shown in Table 4. In Table 4, the real-part reconstruction error of the higher-order derivative is smaller than that of the lower order, but the imaginary part is larger than the lower order. Although in terms of reconstruction error, higher derivatives have no obvious advantages over lower orders, they have a good advantage in terms of sampling rate. This method can be applied effectively when the sampling rate is required to be very low. And the error of reconstruction formula is what we need to study further.

5 Conclusion

In this paper, the reconstruction formulas of the higher-order derivative sampling of band limited signals associated with the FrFT are investigated and derived. Reconstruction formulas include uniform derivative sampling and recurrent nonuniform derivative sampling, which are given in Theorems 1, 2 and 3, respectively. For a bandlimited function $f(t)$ having $n - 1$ order derivative in the FrFT domain, $f(t)$ is determined by uniform sampling points $f^{(l)}(knT)$ ($l = 0, 1, \dots, n - 1$) or nonuniform sampling points $f^{(l)}(n(t_p + kNT))$ ($l = 0, 1, \dots, n - 1$; $p = 1, 2, \dots, N$), in which T is the original sampling interval, and sampling rate of derivative results can be reduced to $\frac{1}{n}$ of original sampling interval. The simulations are performed to verify the correctness of the derived results, the corresponding MSE are also derived. By comparing the MSE and the computational complexity of higher- and lower-order derivative sampling, the advantage of the higher-order derivatives are discussed in details. The

sampling error analysis and high-dimensional sampling problems will be our future research directions.

References

1. L.B. Almeida, The fractional Fourier transform and time-frequency representations. *IEEE Trans. Signal Process.* **42**(11), 3084–3091 (1994)
2. V. Bargmann, On a Hilbert space of analytic functions and an associated integral transform part I. *Commun. Pure Appl. Math.* **14**(3), 187–214 (2010)
3. K.W. Cattermole, The Fourier transform and its applications. *Electron. Power* **11**(10), 357 (2009)
4. I. Daubechies, The wavelet transform, time-frequency localization and signal analysis. *IEEE Trans. Inf. Theory* **36**(5), 961–1005 (1990)
5. Y.C. Eldar, A.V. Oppenheim, Filterbank reconstruction of bandlimited signals from nonuniform and generalized samples. *IEEE Trans. Signal Process.* **48**(10), 2864–2875 (2000)
6. V.D.O. Edwin, J. Renneboog, Some formulas and applications of nonuniform sampling of bandwidth-limited signals. *IEEE Trans. Instrum. Meas.* **37**(3), 353–357 (1988)
7. Q. Feng, B.Z. Li, Convolution theorem for fractional cosine–sine transform and its application. *Math. Methods Appl. Sci.* **40**(10), 3651–3665 (2017)
8. L. Fogel, A note on the sampling theorem. *IRE Trans. Inf. Theory* **1**(1), 47–48 (2003)
9. J.J. Healy, M.A. Kutay, H.M. Ozaktas, J.T. Sheridan, *Linear Canonical Transforms* (Springer, New York, 2016), pp. 241–245
10. Y.C. Jenq, Digital spectra of nonuniformly sampled signals: fundamentals and high-speed waveform digitizers. *IEEE Trans. Instrum. Meas.* **37**(2), 245–251 (1988)
11. D.L. Jagerman, L.J. Fogel, Some general aspects of the sampling theorem. *Inf. Theory Ire Trans.* **2**(4), 139–146 (1956)
12. B.Z. Li, R. Tao, Y. Wang, New sampling formulae related to linear canonical transform. *Signal Process.* **87**(5), 983–990 (2007)
13. Y. Li, Second order derivative sampling and its application into image scaling, in *IEEE International Conference on Communication Software and Networks* (IEEE, 2015), pp. 299–305
14. Y.P. Lin, P.P. Vaidyanathan, Periodically nonuniform sampling of bandpass signals. *IEEE Trans. Circuits Syst. II Analog Digital Signal Process.* **45**(3), 340–351 (1998)
15. D.A. Linden, N.M. Abramson, A generalization of the sampling theorem. *Inf. Control* **3**(1), 26–31 (1960)
16. A.C. McBride, F.H. Kerr, On Namias’s fractional Fourier transforms. *IMA J. Appl. Math.* **39**(2), 159–175 (1987)
17. F. Marvasti, *Nonuniform Sampling: Theory and Practice* (Kluwer Academic/Plenum Publishers, Dordrecht, 2001), pp. 121–156
18. M. Moshinsky, C. Quesne, Linear canonical transformations and their unitary representations. *J. Math. Phys.* **12**(8), 1772–1780 (1971)
19. R.J. Marks, *Advanced Topics in Shannon Sampling and Interpolation Theory* (Springer, Berlin, 1993), pp. 130–131
20. V. Namias, The fractional order Fourier transform and its application to quantum mechanics. *Geoderma* **25**(3), 241–265 (1980)
21. A. Nathan, On sampling a function and its derivatives. *Inf. Control* **22**(2), 172–182 (1973)
22. A.V. Oppenheim, R.W. Schaffer, Discrete-time signal processing. *Electron. Power* **23**(2), 157 (2009)
23. H.M. Ozaktas, M.A. Kutay, Z. Zalevsky, *The Fractional Fourier Transform with Applications in Optics and Signal Processing* (Wiley, London, 2001)
24. H.M. Ozaktas, O. Aytür, Fractional Fourier domains. *Signal Process.* **46**(1), 119–124 (1995)
25. A. Papoulis, Generalized sampling expansion. *IEEE Trans. Circuits Syst.* **24**(11), 652–654 (1977)
26. C.E. Shannon, Communication in the presence of noise. *Proc. Ire.* **86**(1), 10–21 (1998)
27. E. Sejdić, I. Djurović, L. Stanković, Fractional Fourier transform as a signal processing tool: an overview of recent developments. *Signal Process.* **91**(6), 1351–1369 (2011)
28. S. Shinde, V.M. Gadre, An uncertainty principle for real signals in the fractional Fourier transform domain. *IEEE Trans. Signal Process.* **49**(11), 2545–2548 (2001)

29. C.C. Tseng, S.L. Lee, On the reconstruction of derivative sampling method of band-limited signal. *Signal Process.* **129**, 166–182 (2016)
30. R. Tao, B.Z. Li, Y. Wang, Spectral analysis and reconstruction for periodic nonuniformly sampled signals in fractional Fourier domain. *IEEE Trans. Signal Process.* **55**(7), 3541–3547 (2007)
31. G. Xu, X. Wang, X. Xu, The logarithmic, Heisenberg's and short-time uncertainty principles associated with fractional Fourier transform. *Signal Process.* **89**(3), 339–343 (2009)
32. T.Z. Xu, B.Z. Li, *Linear Canonical Transform and its Application* (Science Press, Ottawa, 2013), pp. 665–670
33. J. Yen, On Nonuniform sampling of bandwidth-limited signals. *IRE Trans. Circuit Theory* **3**(4), 251–257 (1956)
34. K. Yao, J. Thomas, On some stability and interpolatory properties of nonuniform sampling expansions. *IEEE Trans. Circuit Theory* **14**(4), 404–408 (1967)
35. Y.N. Zhang, B.Z. Li, ϕ -Linear canonical analytic signals. *Signal Process.* **143**, 181–190 (2018)